Hamiltonian and Lindbladian Simulation of Spin Chains

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I. INTRODUCTION

The Hamiltonian simulation problem can be defined as follows. Given an initial n-qubit state $|\psi(0)\rangle$, a Hamiltonian H and precision ϵ, output the time-evolved quantum state $|\psi(t)\rangle$ such that:

$$\big\| \, |\psi(t)\rangle \; - \; e^{-iHt} \, |\psi(0)\rangle \big\| \; \leq \; \epsilon \tag{1}$$

To simulate e^{-iHt}, it is often convenient to decompose H into a sum of Pauli strings such that $H = \sum_j H_j$ where $H_j = P \in \{I, X, Y, Z\}^{\otimes n}$. This can always be done because $\{I, X, Y, Z\}^{\otimes n}$ forms a basis for the space of n-qubit operators. However, because $[H_i, H_j] \neq 0$, we cannot simply factor $e^{-i(H_i+H_j)t} = e^{-iH_i t} e^{-iH_j t}$. To overcome this, we can use the Lie-Trotter-Suzuki product formula:

$$e^{-i \sum_j H_j \, t} = \lim_{N \to \infty} \Big(\prod_j e^{-iH_j \, \frac{t}{N}} \Big)^N . \tag{2}$$

To implement this on a quantum computer with finite N, one must be able to implement the time evolution circuit for e^{iPt}. Figure 1a illustrates the quantum circuit implementing the unitary e^{iPt}, where P is a weight-3 Pauli string acting nontrivially on qubits i, j, and k where $O \in \{I, H, H_Y\}$ are the basis-rotation gates. In general, once the basis-rotation gates are removed in Figure 1a, the entangling ladder shown implements the unitary $e^{it \, Z_i \otimes Z_j \otimes Z_k}$. Furthermore, by exploiting the basis-change identities $e^{itX} = H \, e^{itZ} \, H$ and $e^{itY} = H_Y \, e^{itZ} \, H_Y$, the circuit structure can realize any Pauli-string evolution e^{iPt} by substituting appropriate basis-rotation gates. This circuit structure can apply to any weight-k Pauli string.

While the Hamiltonian H governs the unitary evolution of a closed system, the dynamics of an open quantum system are described by the Lindblad master equation:

$$\dot{\rho} = \mathcal{L}[\rho] = \underbrace{-\, i[H, \rho]}_{\mathcal{L}_{\text{coh}}[\rho]} + \underbrace{\sum_{i=1}^{S} \Big(L_i \, \rho \, L_i^\dagger - \tfrac{1}{2} \{ L_i^\dagger L_i, \rho \} \Big)}_{\mathcal{D}[\rho]}, \tag{3}$$

where L_i are the jump operators modeling system–environment interactions, and S is their total number. To simulate the full evolution $\rho(t) = e^{t\mathcal{L}}[\rho(0)]$, we can factorize $\mathcal{L}$ into coherent and dissipative evolution and simulate over M discrete steps of size Δt:

$$e^{M \Delta t \, \mathcal{L}} = \Big(\prod_{j=1}^{M} e^{\Delta t \, \mathcal{D}} \circ \mathcal{U}_{\Delta t} \Big), \quad \mathcal{U}_{\Delta t}[\rho] = e^{-i \Delta t H} \, \rho \, e^{i \Delta t H} .$$

Here, $\mathcal{U}_{\Delta t}$ is the usual unitary propagator (implemented with Hamiltonian simulation circuits, requiring its own Trotterization if H's terms do not commute), and the construction of a circuit for the dissipative map $e^{\Delta t \, \mathcal{D}}$ will be addressed in the Methodology.

II. METHODOLOGY

A. Error Analysis and Higher Order Decompositions

The Lie product formula for finite N is an approximation of the time evolution operator. To determine the error analysis of this approximation, we can apply the Baker–Campbell–Hausdorff (BCH) formula:

$$\ln\big(e^X e^Y\big) = X + Y + \tfrac{1}{2}[X, Y] + \tfrac{1}{12}[X, [X, Y]] + \cdots , \tag{4}$$

By substituting $X = -iH_1 \, \Delta t$ and $Y = -iH_2 \, \Delta t$ where $\Delta t = t/N$ into the BCH formula, we find $e^{-iH_1 \, \Delta t} \, e^{-iH_2 \, \Delta t} = e^{-i(H_1+H_2) \, \Delta t} + O(\Delta t^2)$. Since there are $N = t/\Delta t$ such steps, the total error accumulates to $O\big(N \, \Delta t^2\big) = O\big(t \, \Delta t\big) = O\big(t^2/N\big)$. Thus the first-order Trotter decomposition

$$e^{-iHt} \; = \; e^{-i\left(\sum_j H_j\right)t} \; \approx \; \left(\prod_j e^{-iH_j \, \frac{t}{N}} \right)^N , \tag{5}$$

yields an overall error scaling of $O(t^2/N)$. To reduce the leading $O(\Delta t^2)$ error we can introduce the symmetric (Strang) splitting:

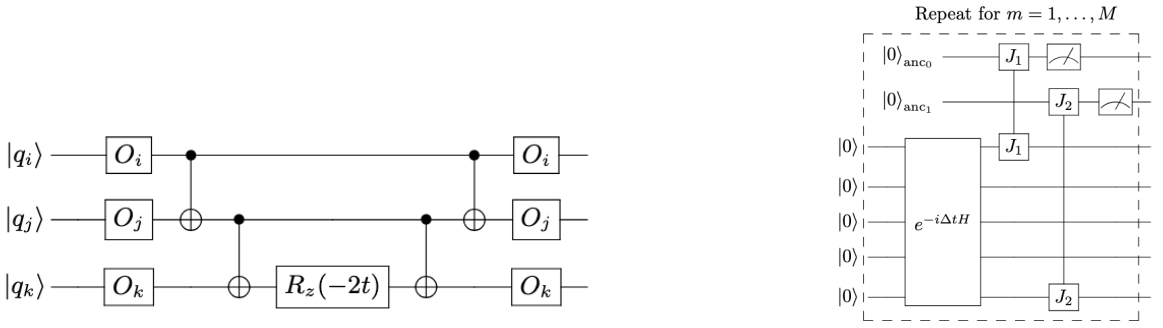
$$S_2(\Delta t) \; = \; \prod_{j=1}^{M} e^{-iH_j \, \frac{\Delta t}{2}} \; \prod_{j=M}^{1} e^{-iH_j \, \frac{\Delta t}{2}} . \tag{6}$$

A similar BCH analysis shows $S_2(\Delta t) = e^{-iH\Delta t} + O(\Delta t^3)$ and hence after N steps , $\big[S_2(t/N)\big]^N = e^{-iHt} + O\big(t^3/N^2\big)$.

More generally, we can build an arbitrarily high even order decompisition S_{2k} by Suzuki's recursive method[1, 2]. We define $p_k = \frac{1}{4 - 4^{1/(2k-1)}}$ and then

$$S_{2k}(\Delta t) = \big[S_{2k-2}(p_k \, \Delta t)\big]^2 \, S_{2k-2}\big((1 - 4p_k) \, \Delta t\big) \, \big[S_{2k-2}(p_k \, \Delta t)\big]^2 . \tag{7}$$

Once again, we can prove via BCH and induction that $S_{2k}(\Delta t) = e^{-iH\Delta t} + O(\Delta t^{2k+1})$, so that the global error after N steps scales as $\big[S_{2k}(t/N)\big]^N = e^{-iHt} + O\big(t^{2k+1}/N^{2k}\big)$.



(a) (b)

FIG. 1: (a) Circuit structure for e^{iPt} for pauli string P. (b) Full Lindbladian simulation circuit of open XXZ model.

Using equations (6), (7) and (8), we can implement trotterized time-evolution circuits of the first order, second order and the $2k$-th order. And using our BCH analyis we can conclude that our error

$$\left\|U^r_{\text{trot}}(N,t)\,|\psi(0)\rangle - e^{-iHt}\,|\psi(0)\rangle\right\| = O\big(t\,(\Delta t)^r\big) = O\Big(\frac{t^{r+1}}{N^r}\Big). \tag{8}$$

for an r-th order trotterization circuit and trotter number N.

B. Hamiltonian Simulation Methods

The transverse-field Ising model (TFIM) is a paradigmatic quantum spin system that captures the competition between nearest-neighbor interactions and a uniform transverse magnetic field. Its Hamiltonian on a chain of N spin-$\frac{1}{2}$ particles is given by

$$H_{\text{TFIM}} = -J\sum_{i=1}^{N-1} Z^{(i)}Z^{(i+1)} + g\sum_{i=1}^{N} X^{(i)}$$

where J is the Ising coupling strength and g the transverse field amplitude. The non-commutativity of the TFIM Hamiltonian prevents an exact factorization of the time-evolution operator e^{-iHt}. We therefore approximate e^{-iHt} using rth-order Trotter–Suzuki decompositions. In this work, we use the TFIM as a benchmark for Trotter analysis and pose the central question: for a given evolution time t and error precision ϵ, which Trotter order r and step count N minimize the total gate count?

To answer this, we define the critical time $t_{\text{crit}}(N,r,\epsilon)$ as the maximal simulation duration for which the Trotter error remains below ϵ. By mapping out $t_{\text{crit}}(N,r,\epsilon)$ across different (N,r) pairs, we can identify the optimal decomposition strategy for any desired precision. To determine $t_{\text{crit}}(N,r,\epsilon)$, we rewrite Equation (8) using the definition of Big-O notation. There exist positive constants $C_r,\ N_0,\ t_0$ such that for all $N \geq N_0$ and $t \geq t_0$,

$$\left\|U^r_{\text{trot}}(N,t)\,|\psi(0)\rangle - e^{-iHt}\,|\psi(0)\rangle\right\| \leq C_r\,\frac{t^{r+1}}{N^r} \tag{9}$$

By computing the exact time-evolved state $e^{-iHt}\,|\psi(0)\rangle$ via matrix exponentiation and comparing it to the Trotterized evolution $U^r_{\text{trot}}(N,t)\,|\psi(0)\rangle$, we can quantify the Trotter error as the ℓ_2-norm distance between the two states. Evaluating this error for various step counts N at a fixed evolution time $t = t_0$ allows us to fit the data and extract the constants C_r. Finally, by equating the bound to the target precision ϵ and solving for t, one obtains the critical evolution time for a given step count N and Trotter order r:

$$t_{\text{crit}}(N,r,\epsilon) = \Big(\frac{N^r\,\epsilon}{C_r}\Big)^{1/(r+1)}. \tag{10}$$

Equivalently, solving the same bound for N yields the minimum number of Trotter steps required to achieve precision ϵ at time t:

$$N_{\min}(t,r,\epsilon) = \Big(\frac{C_r\,t^{\,r+1}}{\epsilon}\Big)^{1/r}. \tag{11}$$

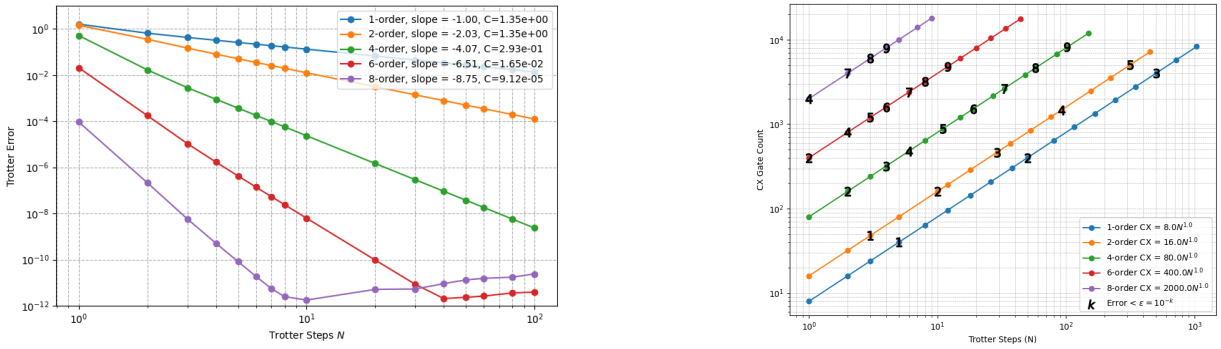
The final task is to determine the total two-qubit gate count of the circuit $U^r_{\text{trot}}(N_{\min},t)$, which can be estimated analytically or extracted by fitting to simulation data. In our implementation, we measure circuit complexity solely by the number of CX gates required.

All quantum circuit simulations in this work were conducted using Qiskit's statevector simulator [3], with the qlib library [4] employed for exact time-evolution calculations.

C. Lindbladian Simulation Methods

The system we will be studying for Lindblad simulation is the open XXZ spin chain governed by the Hamiltonian

$$H_{XXZ} = \sum_{i=1}^{N-1}\big(X^{(i)}X^{(i+1)} + Y^{(i)}Y^{(i+1)} + \Delta\,Z^{(i)}Z^{(i+1)}\big) \tag{12}$$



(a) (b)

FIG. 2: (a) Trotter error vs Trotter steps N for $r = 1, 2, 4, 6, 8$. Fiting used to determine C_r. (b) CX-gate-count scaling for various orders as a function of N. Fitting used to determine G_r.

and jump operators $L_1 = \sqrt{2\epsilon}\,\sigma_+^{(1)}$ and $L_2 = \sqrt{2\epsilon}\,\sigma_-^{(N)}$. In this system, the spin chain interacts with the environment only at its edges.

The main idea of implementing $e^{\Delta t\,\mathcal{D}}$ is to couple the system qubits to two ancilla qubits (one per jump operator), then trace out the ancillas to realize the non-unitary dissipation. Tracing out can be implemented by measuring each ancilla, discarding the outcomes, and resetting the qubits. We perform dissipative evolution by constructing a block operator J from the L_i and implementing $e^{\Delta t\,\mathcal{D}} = e^{-J\sqrt{\Delta t}}$ following the general algorithm for Lindblad simulation adapted from [5]. In our case, the block matrix J is

$$J = J_1 + J_2 = \begin{pmatrix} 0 & L_2^\dagger & L_1^\dagger & 0 \\ L_2 & 0 & 0 & 0 \\ L_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{13}$$

Here the basis for J is the four computational states of the two ancilla qubits, and each block entry applies the corresponding L_i (or $L_i^\dagger$) to the system qubits. For example, L_2 couples $|00\rangle_{\text{anc}}$ and $|01\rangle_{\text{anc}}$ (thereby exciting and relaxing the second ancilla) while applying L_2 or $L_2^\dagger$ to the Nth system qubit, which fully describes the action of J_2. Including both L_i and $L_i^\dagger$ ensures that J is unitary and thus implementable on a quantum computer. Utilizing this block matrix J we can derive equations for J_1 and J_2:

$$J_i = |1\rangle\langle 0|_{\text{anc}^i} \otimes L_i \;+\; |0\rangle\langle 1|_{\text{anc}^i} \otimes L_i^\dagger \tag{14}$$

and, using $\sigma_\pm = \frac{X \pm iY}{2}$, we can derive them in terms of Pauli gates

$$\begin{aligned} J_1 &= \frac{\sqrt{2\epsilon}}{2}\, X^{(\text{anc}_1)} X^{(1)} + \frac{\sqrt{2\epsilon}}{2}\, Y^{(\text{anc}_1)} Y^{(1)}, \\ J_2 &= \frac{\sqrt{2\epsilon}}{2}\, X^{(\text{anc}_2)} X^{(N)} - \frac{\sqrt{2\epsilon}}{2}\, Y^{(\text{anc}_2)} Y^{(N)}. \end{aligned} \tag{15}$$

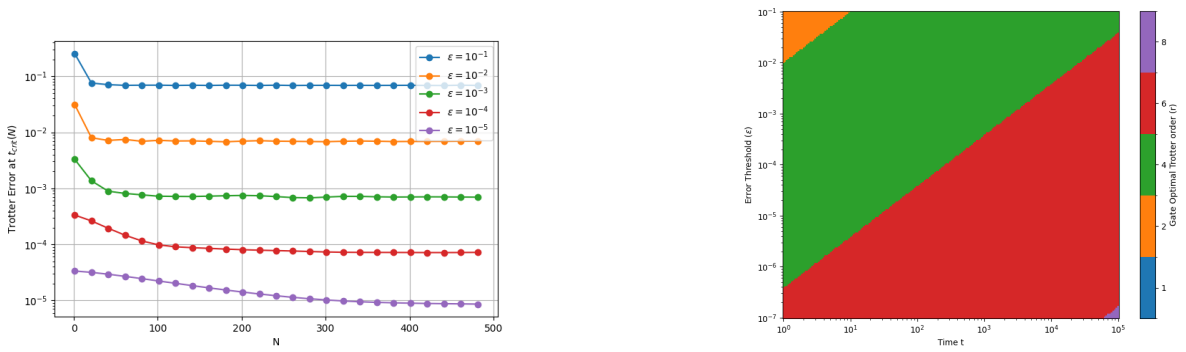
These are simply Pauli strings acting on the system and ancilla qubits, which we know how to implement the time evolution of on our quantum register. Finally, we can construct our circuit for the dissipative evolution according to the following equation:

$$\begin{aligned} e^{\Delta t\,\mathcal{D}}[\rho] = \text{Tr}_{\text{anc}_1,\text{anc}_2}\Big[& e^{-i\sqrt{\Delta t}\,J_1} e^{-i\sqrt{\Delta t}\,J_2} \\ & (|00\rangle\langle 00|_{\text{anc}} \otimes \rho)\, e^{i\sqrt{\Delta t}\,J_2} e^{i\sqrt{\Delta t}\,J_1}\Big] + O(\Delta t^2) \end{aligned} \tag{16}$$

By alternating between dissipative and coherent evolution and simulating over M steps, we construct our full Lindbladian-simulation circuit shown in Figure 1b, where the two-qubit gates connecting the system to our ancilla qubits are $e^{-i\sqrt{\Delta t}\,J_1}$ and $e^{-i\sqrt{\Delta t}\,J_2}$.

III. RESULTS

In our Hamiltonian simulation experiments, we implemented Trotter decompositions of orders $r = 1, 2, 4, 6, 8$, where the higher orders $r = 4, 6, 8$ were generated via Suzuki's recursive formula in equation 8 with base case $r = 2$. We fixed the system parameters to $n = 5$ qubits, $J = 1$, and $g = 1.2$, so that the natural time unit is $t \sim \hbar/J = 1$ (in units where $\hbar = 1$). Additionally, $|\psi(0)\rangle$ is always set to the all-zero state $|\mathbf{0}\rangle$. In our first set of calculations, we evaluated the Trotter error as a function of the number of steps N at evolution time $t = 1$. For each pair (r, N), we constructed the trotterized circuit, simulated the final state using Qiskit's statevector simulator, and computed the $\ell_2 s$-norm error relative to the exact state. The results are displayed in Figure 2a). As predicted by equation 10, $\log \epsilon = -r \log N + \log C_r$. The slopes of the error curves on a log–log plot closely match $-r$, with slight deviations at higher orders due to subleading corrections. From the intercepts of these fits we empirically extract the scaling constants C_r, which will be used to predict t_{crit} and N_{min} in subsequent experiments. Additionally, for higher order Trotterizations we can observe the accuracy reaching machine level precision where accumulated rounding errors start to cross an



(a) (b)

FIG. 3: (a) Error vs N after simulation for $t = t_{\text{crit}}(N, r = 2, \epsilon)$ for various ϵ. (b) Final result of gate-optimal trotter order for each (t, ϵ) point.

increase in error. Data points with error less than 10^{-10} were excluded from fitting.

In the next set of measurements, the CX gate count was determined from the trotterized circuit for each pair (r, N). The results are displayed in Figure 2b in a log–log plot. As expected, the number of CX gates scales linearly with N, since each circuit is repeated in N blocks. The number of CX gates per step, denoted by G_r for each order, is also extracted empirically by fitting, as shown in the plot. For illustrative purposes, we placed an integer marker k along each r-curve at the N value where the error first falls below 10^{-k}. From these markers, we can address our central experimental goal for $t = 1$: finding which r yields the lowest CX count for a given precision $\epsilon = 10^{-k}$. For example, for $k = 1$, $r = 1$ is optimal; for $k = 2$, $r = 2$ is optimal; and for $k = 3, 4, 5, 6$, $r = 4$ is optimal. The next goal is to determine the optimal order over a wide range of simulation times.

Using our values of C_r, we computed the theoretical values for t_{crit}. We then conducted an experiment to verify these predictions. For each tolerance ϵ and each N, we performed an $r = 2$ Trotter simulation up to $t = t_{\text{crit}}$ and calculated the error. If the predicted t_{crit} is correct, the resulting error should approximate ϵ, ideally remaining just under the target. The results, shown in Figure 3a, confirm that for nearly all (N, ϵ) pairs the error is at or just below ϵ. At very low N and high precision, slight deviations appear due to subleading error terms. Similar agreement was observed for other orders r. In summary, our theoretical predictions for both t_{crit} and the corresponding N_{min} are validated.

At this point, all that remains is to compute the minimum number of CX gates for each (r, t, ϵ) according to $CX_{\text{min}}(r, t, \epsilon) = G_r \times N_{\text{min}}(r, t, \epsilon)$. Using the empirically determined constants G_r and C_r, we plotted CX_{min} over a continuous range of t and ϵ. For each (r, ϵ) point, the optimal CX gate count was extracted and displayed as a color map in Figure 3b. Given a desired precision ϵ and simulation time t, this map indicates the gate-optimal Trotter order; substituting that r into the expression for N_{min} then yields the gate-optimal step count. Thus,

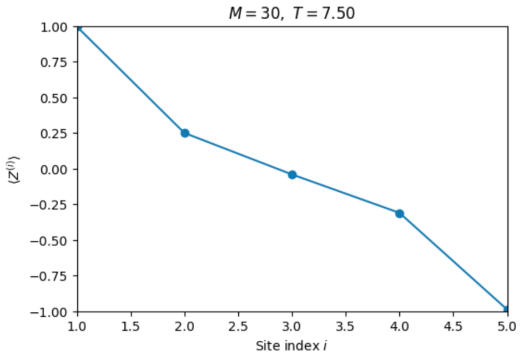


FIG. 4: Spin profile for open XXZ model.

our goal is complete: at low precision and short times, low-order formulas are optimal, whereas at high precision or long times, higher-order decompositions prevail. As a sanity check, the $t = 1$ slice of Figure 3b matches the directly simulated results in Figure 2a, confirming our predictive framework.

In order to benchmark our Lindbladian simulation, we reproduced the spin profiles reported in [6]. That work computes the steady-state expectation values $\langle Z_i \rangle$ of a dissipative XXZ chain using matrix-product states. We implemented the same open-system evolution via our quantum-circuit protocol on $N = 5$ qubits (with anisotropy $\Delta = 0.5$) and ran $M = 30$ Trotter steps. Figure 4 shows the resulting $\langle Z_i \rangle$ profile after the system reaches its non-equilibrium steady state, which closely matches Figure 2 of [6], validating our Lindbladian simulation with quantum computation. Physically, this profile reflects energy leaving through the last qubit and entering from the environment on the first qubit, exactly as expected for boundary-driven dissipation.

REFERENCES

[1] Masuo Suzuki. "Decomposition formulas of exponential operators and Lie exponentials with some applications to quantum mechanics and statistical physics". In: *Journal of mathematical physics* 26.4 (1985), pp. 601–612.

[2] Johann Ostmeyer. "Optimised Trotter decompositions for classical and quantum computing". In: *Journal of Physics A: Mathematical and Theoretical* 56.28 (2023), p. 285303.

[3] Ali Javadi-Abhari et al. *Quantum computing with Qiskit*. 2024. DOI: 10.48550/arXiv.2405.08810. arXiv: 2405.08810 [quant-ph].

[4] Ishaan Ganti. https://github.com/ishaanganti/qlib/tree/main/qlib.

[5] Richard Cleve and Chunhao Wang. "Efficient quantum algorithms for simulating Lindblad evolution". In: *arXiv preprint arXiv:1612.09512* (2016).

[6] Tomaž Prosen. "Exact nonequilibrium steady state of a strongly driven open XXZ chain". In: *Physical review letters* 107.13 (2011), p. 137201.